Pursuant to the DOE Public Access Plan, this document represents the authors' peer-reviewed, accepted manuscript.
The published version of the article is available from the relevant publisher.

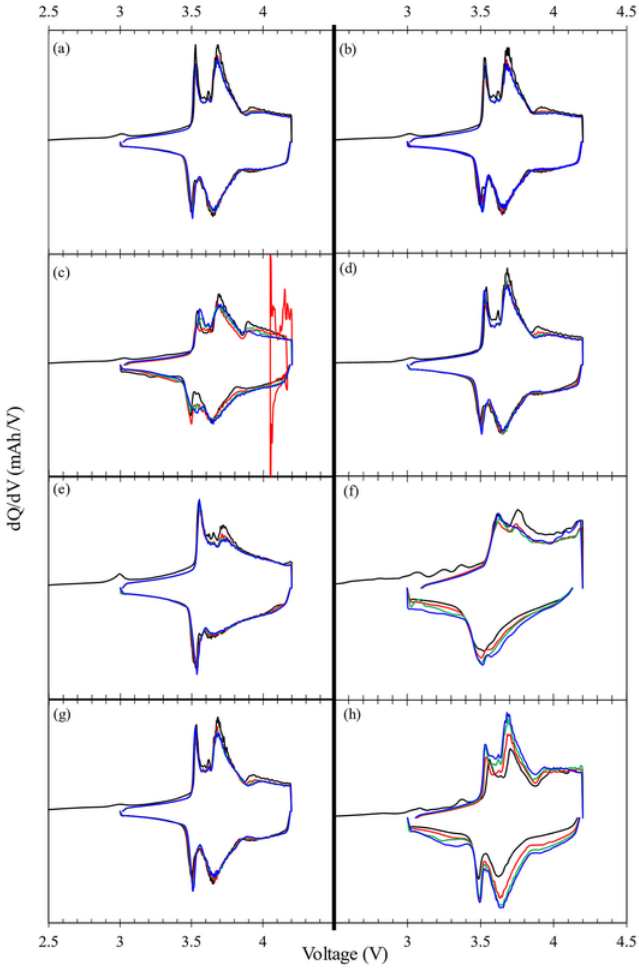

Fig. 3. Differential capacity (dQ/dV) during formation cycles for full cells containing metallic anode impurities: (a) baseline (no impurity); (b) Fe impurity; (c)–(d) Al impurity; (e)–(f) Mg impurity; (g)–(h) Cu impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

contaminated cell; the peak at 0.087 V in the baseline is shifted to 0.072 V; and the peak at 0.056 V in the baseline cell is shifted to 0.042 V. This behavior is limited to the first cycle; at cycle 2 and beyond, the Fe^0-containing anode shows identical electrochemical performance to the baseline cell. Thus, we anticipate that these peaks are associated with irregular SEI formation, likely including the reductive passivation of Fe (e.g., FeO → Fe^0), the dissolution of Li^+ into Fe-containing surface oxides (e.g., $Li_xFe_2O_3$) or weakly catalytic interactions between oxidized Fe^{2+} and electrolyte species. These first cycle losses may contribute to the slight decrease observed in half-cell capacity for Fe^0-containing anodes, relative to pristine anodes (Fig. A3(a)).

In full cells, differential capacity analysis reveals no discernible electrochemical impact on electrochemical performance, as seen in Fig. 3 (b). However, the full-cell specific capacity of cells containing Fe^0 impurity at the anode is reduced relative to the baseline cells, even when the mass of contaminant is omitted from the normalization (Fig. A4(a)). This may be attributed to the irregular first-cycle behavior observed in half cells, which may consume Li^+ through irreversible surface oxide formation. Subsequent surface passivation and the stability of metallic Fe^0 at anodic potentials minimizes further capacity degradation with repeated cycling.

The thermal behavior of full cells containing Fe^0 at the anode is shown in Fig. 4(a). Normalized signal values less than 1 at low potentials (<2.5 V) are attributed to bias error, given the rapid increase of voltage over a brief period of time at the beginning of charge. Thus, the first exothermic signal (~2.25–2.4 V) likely results from Fe^0 oxidation to Fe_2O_3 and Fe_3O_4, both of which occur at low potentials (0.8 V vs Li^+/Li^0) [25]. The observed delay in thermal signal detection is inherent to the kinetics of the process (i.e., heat transfer through the cell, which occurs on the same timescale as the voltage sweep). This error is confined to the initial period of rapid charge, and thus does not propagate. The exothermic peak at ~2.8–3.0 V corresponds to the first-cycle oxidative peak at observed at the same potential in differential capacity analysis (Fig. 3(b)). This electrochemical signal, which is associated with EC reduction to form the SEI [26], is also seen in the first cycle for the baseline cell. However, the fact that a differential exothermic signal is observed *above* the baseline signal for the Fe^0-contaminated cell confirms the participation of Fe^{2+} in irregular SEI formation, and in particular, may suggest a weakly catalytic role of the contaminant in electrolyte reduction. Above 3 V, no difference in thermal signature is observed between the baseline cell and the cell with Fe^0-containing anode.

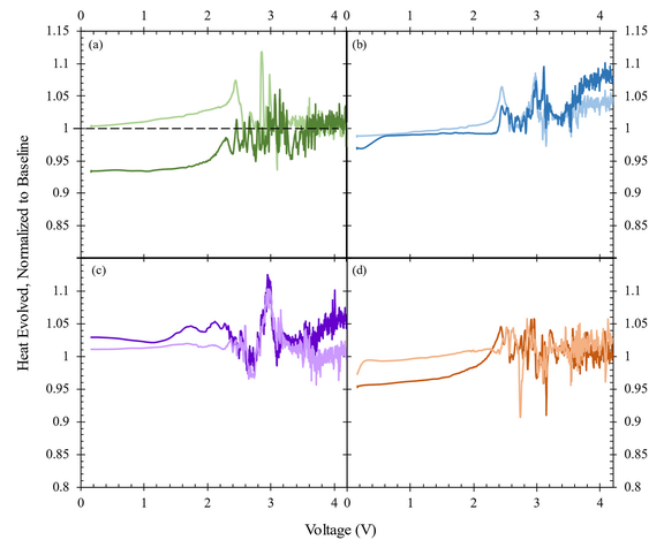

Fig. 4. Thermal signal evolved during first formation charge cycle for full cells containing metallic anode impurities: (a) Fe impurity; (b) Al impurity; (c) Mg impurity; (d) Cu impurity. For each contaminated cell, thermal signal has been normalized to the signature of a representative baseline cell – such that the baseline cell has a normalized value of 1 across all voltages (i.e., the normalized baseline thermal signal can be represented as the dashed black line in (a)). For graphical clarity, the normalized baseline signal has been omitted from plots (b)–(d).

3.2.2. Al^0-contaminated anode

Electrochemical reactivity of a metallic Al^0 impurity with the anode is observed in both half-cell and full-cell format, consistent with the oxophilic [27] and Li-reactive behavior of Al^0 at low potentials. As shown in Fig. 2(c), new dominant oxidative and reductive peaks emerge at 0.027 V during the first cycle; this may be attributable to the evolution of Al–Li Zintl phases, which are anticipated to form at ~400 mV. Notably, this phase formation is only found to occur during the first formation cycle, implying either full consumption of Al^0 impurity with electronic isolation (i.e. trapped in SEI layer) or a passivation of the Al^0 surface, inhibiting further reactivity. Additionally, the first-cycle reductive peak present at 0.167 V in the baseline cell shifts to 0.143 V in the Al^0-containing system as compared to the baseline. The observed downward shift in reduction potential supports the notion of participation of precipitated Al species in the SEI, which may impede intercalation at surface sites.

Pursuant to the DOE Public Access Plan, this document represents the authors' peer-reviewed, accepted manuscript. The published version of the article is available from the relevant publisher.

The impact of Al^0 impurity is also manifest in full-cell format for a subset of the cell replicates tested. The noise observed at high potential in Fig. 3(c) during the second formation cycle may reflect electrolyte reactivity, and more specifically, increased reactivity associated with the presence of the Al contaminant. Reduced Al species will react with solvents analogously to other main group elements (e.g., Si), and therefore additional reactivity associated with electrolyte decomposition and SEI formation is anticipated for Al^0-contaminated cells. Additionally, upon discharge, a small peak at 3.54 V emerges in the cell with Al^0 impurity. Based on previously-reported deconvolutions [23], it is probable that this peak is associated with the anode, and is masked in the baseline cell as a weak shoulder in the dominant reductive peak at 3.66 V. In the cell with Al^0-containing anode, this reductive peak is found to shift to a lower potential, again implying the disruptive nature of Al^0 on anode surface stabilization.

Notably, a second full-cell replicate containing an Al^0-contaminated anode showed no discernible electrochemical differences as compared to the baseline (Fig. 3(d)), suggesting a batch-scale heterogeneity in contaminant dispersion. Such heterogeneity is anticipated in industrial recycled materials, and thus capturing cell-to-cell variation is critical to accurately representing the impacts of contaminants on recycled material outputs.

As with the Fe^0-contaminated anodes, the Al^0-contaminated anodes generally show a slight decline in specific capacity relative to the baseline material, in both half cell (Fig. S3(b)) and full cell (Fig. A4(b)) format. This suggests irreversible Li^+ loss, which may be attributable to irreversible first-cycle Li–Al Zintl phase formation, side reactions that consume lithium cations, or subtle changes to the SEI composition induced by the aluminum.

The thermal signature of Al^0-contaminated full cell, shown in Fig. 4 (b), is comparable to that of Fe^0-contaminated cells. Notable exothermic peaks occur at ~2.4 V, attributable to Al^0 oxidation (occurring at 1.38 V [28], but delayed in detection due to inherent voltage bias, as discussed in Section 3.2.1) and ~2.8 V, suggesting participation of the oxidized Al species in the EC reduction leading to SEI formation. Notably, the dual peaks at ~2.8 V and ~3.1 V may correspond to reaction associated with the two predominant oxidized forms of Al, namely, Al^{2+} (AlO) and Al^{3+} (Al_2O_3). High-potential thermal signatures correspond neatly to differential electrochemical behavior in this voltage range (Fig. 3(c)), suggesting more highly exothermic electrolyte reactivity in the Al^0-contaminated cells as compared to the baseline cells. It is noted that, although Al^0 is present in the baseline cell (as cathode current collector), the current collector foil is ostensibly pre-passivated with a stabilizing layer of Al_2O_3 due to atmospheric oxidation. Thus, the inclusion of practical contamination from pre-passivated current collectors – rather than pure Al^0 powder – may alter the observed thermal signature; this should be evaluated in future work.

3.2.3. Mg^0-contaminated anode

Half-cell analysis of Mg^0-containing anodes, given in Fig. 2(d), suggests phase formation behavior at low potential (~0.028 V), similar to that observed for anodes with Al^0 impurity. However, in contrast to Al^0-contaminated anodes, the Mg^0-contaminated anodes show continuous reactivity throughout the four formation cycles. Even in the presence of an infinite Li reservoir (i.e., half cell format), such Mg–Li alloying appears to significantly impact cell capacity (Fig. A3(c)), ostensibly through reduced ionic diffusion at the metallic Li surface. The impact on full cell capacity (Fig. A4(c)) is heterogeneous by replicate, with two of the four cell replicates showing drastically reduced capacity. This is consistent with an irreversible loss of cyclable Li^+ due to alloying reactions with Mg.

Half cells containing Mg^0 impurities also show increased hysteresis in the dominant redox peak following the first formation cycle. In particular, the oxidative peak is found to shift from 0.109 V in the baseline cell to 0.120 V in the cell with Mg^0-contaminated anode, with no evident change in the position of the reductive peak. Increased hysteresis indicates increasing electrode impedance, which could be linked to the formation of diffusion-inhibiting Li–Mg alloys. This may also alter the practical potential experienced by the cathode. Finally, the first-cycle reductive peak at 0.167 V in the pristine material shifts to 0.136 V in the Mg^0-contaminated material, suggesting a more thermodynamically unfavorable passivation.

In full cells, a significant decline in intensity of the dominant redox peak at 3.65 V is observed for all cell replicates (Fig. 3(e)–(f)), particularly upon discharge (reduction). This redox peak is typically associated with cathodic behavior [23], supporting the notion that the increased hysteresis induced by the Mg^0 contaminant may be impacting the true cathodic potential. Additionally, a new peak is observed at high potentials (~4.2 V), suggesting increased electrolyte reactivity in the presence of oxidized Mg species.

Slight variability in thermal signature is observed between replicates of Mg^0-containing cells. Specifically, one cell shows low-potential exothermic behavior (ostensibly Mg^0 oxidation to MgO), while the other does not; the former cell also shows increased exothermic signal at high potentials. This cell-to-cell heterogeneity may be associated with local differences in contaminant size and loading, and reflects the range of behaviors that may be observed in cells fabricated from a true end-of-life black mass. The predominant thermal peak for both cells occurs at ~2.75 V, and is likely tied to EC reduction – implying the accelerating impact of Mg (and likely Mg–Li alloy) on electrolyte decomposition and associated SEI formation.

3.2.4. Cu^0-contaminated anode

The behavior of anodes containing Cu^0 impurities is highly erratic during formation cycling in half-cell format, as shown in Fig. 2(e), with dQ/dV peak position varying significantly between cycles. The first formation cycle shows behavior similar to the baseline material; this is anticipated, as Cu is not expected to react with Li at anodic potentials. However, in the subsequent formation cycles, several erratic spikes in differential capacity are observed, implying that the metallic Cu^0 impurity is not entirely inert.

In full cells, the impact of Cu^0 contamination at the anode is heterogeneous between replicates. Some cells show no observable electrochemical difference from the baseline (Fig. 3(g)), but others show a trend of *increasing* dQ/dV intensity during formation (i.e., from cycle 1 to cycle 4; Fig. 3(h)). This trend is also manifest in full-cell capacity analysis (Fig. A4(d)), where two of the four replicates are found to show increased capacity relative to the baseline after extended cycling. We suggest that Cu impurities may act as a conductive additive when sited at the anode, increasing electrode conductivity and thereby enhancing cell performance. However, given that this effect is not observed uniformly between cell replicates, further study should be undertaken to further probe the source of this behavior.

Finally, increased reactivity at high potentials (~4.2 V) is observed for Cu^0-containing anodes in full cells, similar to that seen for Mg_0-contaminated anodes. This may be due to disruption of surface sites by oxidized Cu species (CuO, Cu_2O), or enhanced reactivity induced by ionic Cu at high potential.

Minimal to no impact on thermal signature is observed for the Cu^0-containing full cells, as compared to the baseline cell. This is physically reasonable: Given that Cu is already present in the baseline cell (as current collector), reactivity from Cu oxidation would be captured in the baseline signature; the lack of an additional peak at ~2.8 V, as has been observed for the other anode contaminants, implies a minimal impact of Cu on SEI formation.

3.2.5. Summary of anode contaminant behavior

From a cell performance perspective, Cu^0 and Mg^0 are found to be the most disruptive anode contaminants in both half-cell and full-cell format. Predominant modes of reactivity observed for anode contami-

Pursuant to the DOE Public Access Plan, this document represents the authors' peer-reviewed, accepted manuscript.
The published version of the article is available from the relevant publisher.

nants include alloying and Zintl phase formation at low potentials, and catalytic acceleration of electrolyte decomposition induced by the high oxophilicity of the contaminants under study. The observed reduction in long-term cycling capacity, particularly for cells with Cu^0 and Mg^0 at the anode, is attributed to an increase in the irreversible consumption of Li during initial formation cycling, which appears to form an inefficient and resistive SEI at the anode that inhibits subsequent cycling performance. For all anode contaminants except Cu^0, analysis of thermal reveals the role of the oxophilic impurities in enhancing electrolyte degradation, which is not immediately evident through electrochemical analysis alone. Finally, there is no evidence of physical penetration of the separator (i.e., hard internal short circuit) for any of the metallic contaminants.

3.3. Impact of metallic impurities on cathode performance

As for the contaminated anodes described in **Section 3.2.**, the electrochemical and thermal signatures of contaminated cathodes has been evaluated using a combination of half-cell and full-cell analysis. The cathode impurities studied include Fe, Al, Mg, Cu, and Si. The disparity in anodic and cathodic operating potential leads to differing impacts on cell performance when these contaminants are sited in the cathode, rather than in the anode, and such distinction is crucial to understanding the range of impacts associated with metallic contaminants introduced during recycling.

Differential capacity plots (dQ/dV) of half cells with impurity-containing cathodes, as compared to pristine cathodes, are shown for the first four formation cycles in Fig. 5; analogous dQ/dV plots for full cells with impurity-containing cathodes are shown for formation cycles in Fig. 6. The electrochemical performance of cells containing cathode impurities was more consistent between cell replicates than the performance of cells containing anode impurities, and thus, a single representative dQ/dV plot is shown for each cell condition. While emphasis will again be placed on electrochemical signatures during formation, the capacities of half and full cells containing cathode contaminants over the course of extended cycling are provided in the Appendix (Figs. A6-A7).

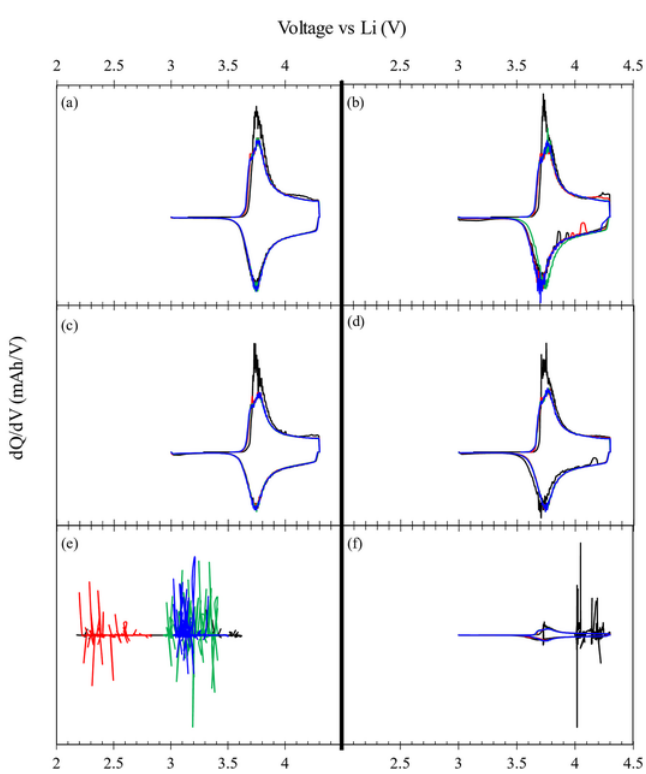

Fig. 5. Differential capacity (dQ/dV) during formation cycles for cathode half cells (NMC-111 vs Li^0) prepared with metallic impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity; (f) Si impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

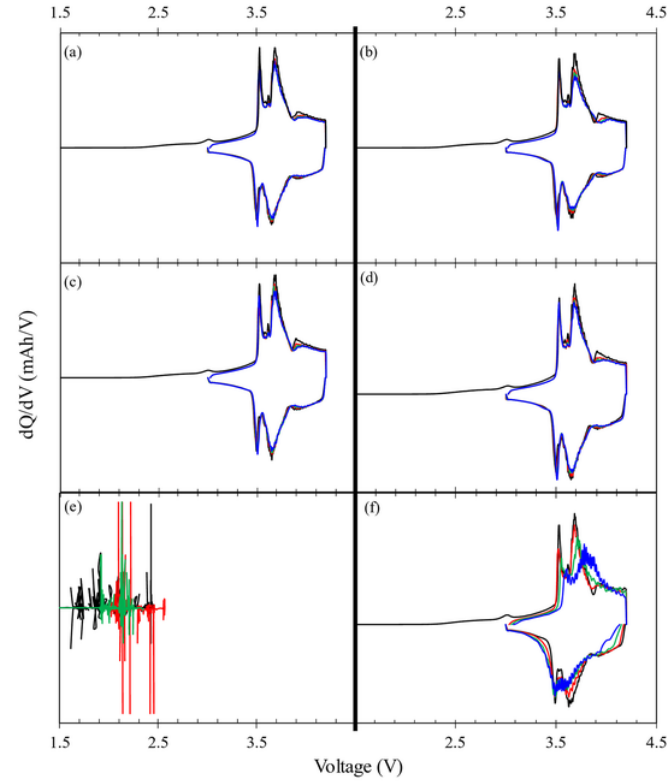

Fig. 6. Differential capacity (dQ/dV) during formation cycles for full cells containing metallic cathode impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity; (f) Si impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Thermal signals for full cells with metallic impurities at the cathode are presented in Fig. 7. As in Fig. 4, the signals have been normalized by the signature of a baseline cell to highlight the distinguishing features attributable to the impurity. Raw thermal signals for each cell are given in Fig. S8.

Each metallic cathode impurity will be discussed in the following sections.

3.3.1. Fe^0-contaminated cathode

Fe is anticipated to be electrochemically and chemically stable in the cathodic voltage range, and thus a minimal impact on electrochemical performance would be expected. In full-cell format, no noticeable change in electrochemical signature is observed for the Fe^0-contaminated cathode, as compared to the baseline material (Fig. 6(b)). However, in half cells, differential capacity plots show irregularity during the highest voltage portion of charging (oxidation) and the high-voltage portion of discharge (reduction) during the two initial formation cycles (Fig. 5(b)). Interestingly, this corresponds to the voltage regime in which the most significant heat is evolved from Fe^0-contaminated full cells, as shown in Fig. 7(a). This implies a weakly catalytic role of the oxidized Fe^0 impurity in electrolyte decomposition, which occurs at high potentials. The fact that such reactivity in full cells is observable via thermal analysis, but not via electrochemical analysis,

Pursuant to the DOE Public Access Plan, this document represents the authors' peer-reviewed, accepted manuscript.
The published version of the article is available from the relevant publisher.

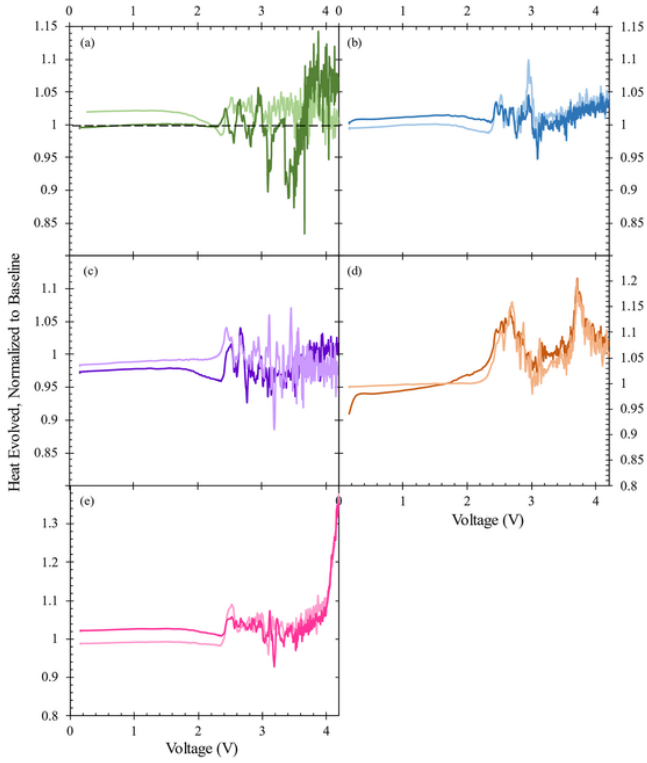

Fig. 7. Thermal signal evolved during first formation charge cycle for full cells containing metallic cathode impurities: (a) Fe impurity; (b) Al impurity; (c) Mg impurity; (d) Cu impurity; (e) Mg impurity. For each contaminated cell, thermal signal has been normalized to the signature of a representative baseline cell – such that the baseline cell has a normalized value of 1 across all voltages (i.e., the normalized baseline thermal signal can be represented as the dashed black line in (a)). For graphical clarity, the normalized baseline signal has been omitted from plots (b)–(e).

highlights the complementary role of these two techniques in identifying the presence of metallic contaminants.

A more significant capacity decline is observed for full cells containing Fe^0 contaminant in the cathode than for half cells with the Fe^0 impurity (Fig. A6(b), Fig. A7(b)). This observed decrease in capacity stands in contrast to the evident lack of reactivity suggested by differential capacity analysis, and supports the notion of Fe^0 playing a catalytic role in electrolyte decomposition pathways and increasing irreversible Li^+ consumption during the initial formation cycles.

3.3.2. Al^0-contaminated cathode

In half cell format, cells containing Al^0 impurity show novel oxidative peaks at 3.96 V and 4.0 V during the first formation cycle (Fig. 5(c)); there is also a slight first-cycle oxidative peak increase at the highest potentials (~4.2–4.3 V). While the formation of Al_2O_3 is theoretically anticipated to occur at relatively low potentials (~1.39 V vs Li^+/Li [29]), we suggest that the oxygen required for Al oxidation may be evolved during electrolyte decomposition, which does not occur until higher potentials. Further, it has been reported that the Al_2O_3 may subsequently react with the fluorinated electrolyte salt ($LiPF_6$) to fluorinate the aluminum surface layer [30]. We suggest that the first novel oxidative peak (3.96) may be associated with Al_2O_3 passivation once the electrolyte has decomposed to produce free oxygen, while the second oxidative peak (4.0 V) may be associated with the fluorination of the Al_2O_3 passivation layer through reaction with HF (produced from decomposition of the alkali electrolyte salts) [30]. Following this first-cycle passivation, the Al^0 contaminant appears to be electrochemically inert.

Minimal to no impact is observed from cathode-sited Al^0 impurity on the full cell differential capacity signature (Fig. 6(c)), and the capacity of full cells containing Al^0-contaminated cathodes is on par with the baseline cells during extended cycling (Fig. A7(c)). The thermal signatures for cells contaminated with Al^0 at the cathode show a spike in heat evolved at ~3.0 V, followed by a gradual rise in heat generation at higher potentials (Fig. 7(b)). It is notable that this thermal signal evolves above the signal of the baseline cell, suggesting specific reactivity of the Al^0 contaminant, rather than the cathode current collector. This signal does not directly align with features observed in the electrochemistry, and thus the mechanism of reactivity leading to this heat generation is not immediately apparent. Taken together, the observed thermal and electrochemical behavior for Al^0-contaminated cathodes implies Al^0 reactivity under cathodic conditions, but with a minimal impact on practical full-cell electrochemical performance.

3.3.3. Mg_0-contaminated cathode

As for Al^0 contaminants, Mg^0 impurities show minimal impact on half-cell cathode performance (Fig. 5(d)), and no distinguishable impact on full-cell electrochemical signature (Fig. 6(d)). However, there are critical differences in the half-cell differential capacity plots that indicate a varying mode of reactivity between these two contaminants. In particular, as seen in Fig. 5(d), Mg^0-contaminated cathodes show irregular reductive behavior at 4.16 V during the first cycle, and a shift to lower voltage in the first-cycle dominant reductive peak, from 3.78 V in the baseline material to 3.70 V in the Mg^0-containing materials, is observed. As has been suggested for Mg^0 contamination at the anode, it is probable that the Mg^0 contaminant may be impacting the practically manifest cathodic potential, leading to the observed increase in hysteresis of the dominant redox peak. Additionally, a new peak is observed at high potentials (~4.2 V), suggesting increased electrolyte reactivity in the presence of oxidized Mg species.

Given the moderately high oxophilicity of Mg^0, it is probable that a stable and passivating film of MgO_2 may form on the Mg^0 surface, with oxygen evolved from electrolyte reduction. This leads to the complete passivation of the Mg^0 contaminant after the first cycle, resulting in minimal disruption of electrochemical performance. The lack of practical impact of the Mg^0 contaminant is evident in both the thermal signature (Fig. 7(c)) and electrochemical capacity (Fig. A7(c)) of full cells containing Mg^0-contaminated cathodes.

3.3.4. Cu_0-contaminated cathode

At cathodic potentials, metallic Cu^0 is highly unstable; this is manifest as a significant disruption in the electrochemical signatures of both half cells (Fig. 5(e)) and full cells (Fig. 6(e)). At potentials around 2.8 V, Cu^0 will oxidize to Cu^+, which is soluble in electrolyte; at higher potentials (around 3.7 V), Cu^+ will further oxidize to soluble Cu^{2+}. These two oxidation potentials are clearly observed in the thermal profiles of full cells contaminated with Cu^0 at the cathode: As shown in Fig. 7(d), exothermic peaks are evident between ~2.4–2.8 V ($Cu^0 \rightarrow Cu^+$) and at 3.7 V ($Cu^+ \rightarrow Cu^{2+}$). The formation of soluble ionic species enables diffusion through the separator, which may lead to crossover (i.e., cathode contaminant deposition at the anode) and may also induce a hard internal short through the formation of lithium dendrites. In full cells, it is anticipated that ionized Cu^+ and Cu^{2+} migrating to the anode may incorporated into the SEI. This is anticipated to induce a deleterious effect on SEI growth, leading to both excessive Li consumption and the formation of a resistive passivating layer that inhibits Li-ion transport. As such, an extreme reduction in cyclable capacity is observed for full cells containing metallic Cu^0 (Fig. A7(e)).

While the erratic differential capacity signature observed for Cu^0-contaminated half cells in Fig. 5(e) suggests an internal short, an interesting capacity recovery is observed during subsequent cycling (Fig. A6(e)). Following what appears to be a short during initial formation cycles, half cells containing Cu^0 in the cathode then show only a slightly

Pursuant to the DOE Public Access Plan, this document represents the authors' peer-reviewed, accepted manuscript. The published version of the article is available from the relevant publisher.

depressed capacity relative to the baseline throughout subsequent cycling. The capacity drop towards the end of cycling implies a restriction on the functionally infinite Li reservoir, perhaps suggesting an early deposition and passivation of Cu on the Li surface. This would lead to regions of "dead" Li, which could no longer participate in electrochemical reaction. The observed capacity recovery in half cells containing Cu^0 impurity at the cathode runs counter to previous reports on the impacts of dissolved copper in Li-ion half cells [11,25], which have suggested that ionic Cu^+/Cu^{2+} is deleterious to half cell operation.

3.3.5. Si^0-contaminated cathode

As shown in Fig. 5(f), cathodes containing Si^0 impurity show evidence of significant first-cycle reactivity at high potentials (above 4.0 V) in half-cell format. This reactivity is largely limited to the first formation cycle, with half-cell differential capacity signatures stabilizing after the first cycle. This behavior may be attributed to Si^0 reactivity with the electrolyte, and in particular, the evolution of oxygen that reacts with Si to form silicon-oxide species (e.g., SiO_2), which may subsequently react with lithium to form electrochemically stable Li_2SiO_3. In the presence of an infinite Li^0 reservoir (i.e., half cell), this loss of lithium does not significantly impact capacity performance in cells containing Si^0-contaminated cathodes (Fig. A6(f)). However, in full cells, a significant decline in capacity is observed (Fig. A7(f)). Notably, the capacity of cells with Si^0 impurities at the cathode remains stable throughout cycling, with the entirety of the capacity loss occurring within the first 10 cycles. This finite reactivity may be linked to the initial quantity of impurity present, and suggests that cathode-sited Si^0 impurity will continue to react with electrolyte-derived oxygen and Li until the Si_0 has been fully passivated to Li_2SiO_3.

The thermal signatures for full cells with Si^0 contaminant at the cathode (Fig. 7(e)) align well with the proposed mode of electrochemical reactivity; namely, the strong exothermic reactivity above 3.5 V may be tied to the formation of SiO_2, and then subsequently Li_2SiO_3.

Differential capacity analysis of full cells with Si^0-contaminated cathodes (Fig. 6(f)) suggests a shift to higher potentials of the two dominant oxidative redox peaks with formation cycling (i.e., from cycles 1 to 4) and a concurrent increase in hysteresis and decline in the intensity of both dominant redox peaks. This implies a growth in impedance, ostensibly due to the deposition of evolved Li_2SiO_3 on the anode.

3.3.6. Summary of cathode contaminant behavior

In terms of practical cell performance, Cu^0 and Si^0 impurities at the cathode most significantly impact long-term reversible cycling capacity. For all other contaminants, effective passivation occurs during the first cycle, leading to a gravimetric capacity penalty but minimal additional impact on cell performance. Notably, none of the cathode contaminants are found induce a hard internal short (e.g., through separator penetration) under the studied conditions. Crossover is anticipated in the case of the Cu^0 impurity at the cathode, whereas the mode of reactivity for the Si^0 impurity is more likely linked to a continuous passivation over numerous cycles. Thermal signatures neatly complement the observed electrochemical reactivity in most cases; however, for the Fe^0 contaminant, additional thermal signals are observed that cannot readily be paired with electrochemical signatures. This highlights the need for complementary analysis and may also direct further study into the mechanism of Fe^0 reactivity under cathodic conditions.

CRediT authorship contribution statement

Kae E. Fink: Writing – original draft, Revised Draft, Responses to Reviewers, Formal analysis, Visualization. **Bryant J. Polzin:** Conceptualization, Methodology, Writing – original draft, Funding acquisition, Project administration. **John T. Vaughey:** Formal analysis. **Joshua J. Major:** Investigation, Formal analysis, Software. **Alison R. Dunlop:** Investigation, Data curation, Visualization. **Stephen E. Trask:** Investigation. **Gerald T. Jeka:** Investigation. **Jeffrey S. Spangenberger:** Supervision. **Matthew A. Keyser:** Conceptualization, Methodology, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was performed in part through the ReCell Center, which gratefully acknowledges support from the U.S. Department of Energy (DOE), Office of Energy Efficiency and Renewable Energy, and the Vehicle Technologies Office. Contribution by Argonne National Laboratory (ANL) is based upon work supported by Laboratory Directed Research and Development (LDRD) funding from ANL, provided by the Director, Office of Science, of the U.S. Department of Energy under Contract No. DE-AC02-06CH11357. Contribution by the National Renewable Energy Laboratory (NREL) is supported by the Alliance for Sustainable Energy, LLC, for the U.S. Department of Energy (DOE) under Contract No. DE-AC36-08GO28308. The views expressed in this article do not necessarily represent the views of the DOE or the US government. The US government and the publisher, by accepting the article for publication, acknowledge that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for US government purposes.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2021.230760.

References

[1] A. Gupta, N. Paranjape, Lithium Ion Battery Market Outlook 2020-2026, Global Market Insights, 2020. https://www.gminsights.com/industry-analysis/lithium-ion-battery-market, accessed December 22, 2020.

[2] European Commission, Joint Research Centre., Li-Ion Batteries for Mobility and Stationary Storage Applications: Scenarios for Costs and Market Growth, Publications Office, LU, 2018. https://data.europa.eu/doi/10.2760/87175. (Accessed 22 December 2020).

[3] K. Richa, Sustainable Management of Lithium-Ion Batteries after Use in Electric Vehicles, Rochester Institute of Technology, 2016. https://scholarworks.rit.edu/cgi/viewcontent.cgi?article=10463&context=theses.

[4] M. Chen, X. Ma, B. Chen, R. Arsenault, P. Karlson, N. Simon, Y. Wang, Recycling end-of-life electric vehicle lithium-ion batteries, Joule 3 (2019) 2622–2646, https://doi.org/10.1016/j.joule.2019.09.014.

[5] F. Peng, D. Mu, R. Li, Y. Liu, Y. Ji, C. Dai, F. Ding, Impurity removal with highly selective and efficient methods and the recycling of transition metals from spent lithium-ion batteries, RSC Adv. 9 (2019) 21922–21930, https://doi.org/10.1039/C9RA02331C.

[6] Y. Zhao, X. Yuan, L. Jiang, J. Wen, H. Wang, R. Guan, J. Zhang, G. Zeng, Regeneration and reutilization of cathode materials from spent lithium-ion batteries, Chem. Eng. J. 383 (2020) 123089, https://doi.org/10.1016/j.cej.2019.123089.

[7] T. Or, S.W.D. Gourley, K. Kaliyappan, A. Yu, Z. Chen, Recycling of mixed cathode lithium-ion batteries for electric vehicles: current status and future outlook, Carbon Energy 2 (2020) 6–43, https://doi.org/10.1002/cey2.29.

[8] M. Eilers-Rethwisch, M. Winter, F.M. Schappacher, Synthesis, electrochemical investigation and structural analysis of doped Li[Ni0.6Mn0.2Co0.2-xMx]O2 (x = 0, 0.05; M = Al, Fe, Sn) cathode materials, J. Power Sources 387 (2018) 101–107, https://doi.org/10.1016/j.jpowsour.2018.02.080.

[9] R. Zhang, Z. Meng, X. Ma, M. Chen, B. Chen, Y. Zheng, Z. Yao, P. Vanaphuti, S. Bong, Z. Yang, Y. Wang, Understanding fundamental effects of Cu impurity in different forms for recovered LiNi0.6Co0.2Mn0.2O2 cathode materials, Nano Energy 78 (2020) 105214, https://doi.org/10.1016/j.nanoen.2020.105214.

[10] Q. Sa, J.A. Heelan, Y. Lu, D. Apelian, Y. Wang, Copper impurity effects on LiNi1/3Mn1/3Co1/3O2 cathode material, ACS Appl. Mater. Interfaces 7 (2015) 20585–20590, https://doi.org/10.1021/acsami.5b04426.

[11] S. Park, D. Kim, H. Ku, M. Jo, S. Kim, J. Song, J. Yu, K. Kwon, The effect of Fe as an impurity element for sustainable resynthesis of Li[Ni1/3Co1/3Mn1/3]O2 cathode material from spent lithium-ion batteries, Electrochim. Acta 296 (2019)